

Wilson Wu

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Portfolio: wilsonwu2007.github.io

EDUCATION

University of California, Los Angeles

Los Angeles, CA

Bachelor of Science in Electrical Engineering

- **Honors:** ECE Fast Track Honors Program (Top 20)
- **Relevant Coursework:** Intro to Electrical Engineering, Internet of Things, Electricity and Magnetism, Object-Oriented Programming (C++), Data Structures and Algorithms

TECHNICAL SKILLS

Design Tools: KiCad, PSpice, LTSpice, OnShape

Test Equipment: Oscilloscope, Bench Power Supply, Digital Multimeter (DMM), Soldering Iron

Software Tools: Arduino IDE, Git, VS Code

Languages: C/C++, Python, Java, HTML/CSS, JavaScript, Dart

EXPERIENCE

Bruin Supermileage

University of California, Los Angeles

Powertrain Project Engineer

June 2026 – Present

- Designing a brake chopper board to protect the DC-bus from voltage spikes during regenerative braking, switching excess energy into an off-board burn resistor before reaching the 60V competition hard limit
- Defined system requirements through research into powertrain architecture and competition rules, specifying a hysteresis trigger band, 70V component rating, 3200W power dissipation target, and sub-2 microsecond switching speed

Low Voltage Engineer

October 2025 – June 2026

- Designed and validated a 2-layer KiCad PCB implementing a multi-sheet fail-safe schematic in KiCad protecting the car's hydrogen fuel cells and motor controller, implementing fault logic for hydrogen leak detection, ignition interlock, and driver e-stop buttons
- Engineered a single-fault tolerant hardware interlock combining hydrogen sensor inputs, ignition signal inputs, and e-stop button isolation to guarantee power isolation independent of semiconductor fault modes
- Validated system functionality through bench testing including fault isolation across all e-stop and sensor fault conditions, verifying relay actuation via multimeter and bench power supply prior to competition

UCLA Plasma Accelerator/Laser-Plasma Interactions Group

University of California, Los Angeles

Undergraduate Researcher

June 2026 – September 2026

- Investigated auto-resonant beat-wave plasma excitation, implementing numerical differential models to simulate nonlinear wave amplitude growth and validating results against published literature
- Configured and ran particle-in-cell (PIC) simulations using OSIRIS, tuning grid resolution and particle count parameters to balance simulation accuracy and signal quality across multiple test scenarios
- Developed Python data pipeline to process large HDF5 simulation outputs using FFT spectral analysis, Hilbert transform envelope detection, and phase-space visualization to extract wave dynamics

PROJECTS

Gacha Thrift | *ESP32-S3, C++, Python*

- Built a gacha-style thrift clothing dispenser in 36 hours, integrating mechanical, electrical, and software subsystems from scratch into a functional end-to-end product
- Implemented closed-loop DC motor control on an ESP32-S3 using encoder interrupt feedback and a trapezoidal speed ramp for precise 90° carousel indexing per dispense
- Debugged hardware timing issues in the dispensing mechanism through iterative mechanical adjustment and encoder parameter tuning until consistent dispense behavior was achieved
- Synced physical capsule dispense with a browser gacha animation via a Python WebSocket bridge reading ESP32 serial output in real time

Micromouse | *KiCad, STM32, C Programming Language*

- Led a team designing an autonomous maze-solving robot, coordinating hardware and firmware development across power, sensing, and motor control subsystems
- Designed a mixed-signal PCB layout integrating 4 IR emitter/detector pairs and a gyroscope for sensing, physically routing low-amplitude sensor traces from high-current motor driver traces to protect signal integrity
- Identified post-fabrication layout errors and resolved through jumper wire patches and STM32 firmware corrections, maintaining full robot functionality
- Developed an H-bridge motor driver circuit for independent left/right DC motor control, accounting for shoot-through protection and flyback protection to protect against back-emf
- Implementing a flood-fill maze-solving algorithm in C on an STM32 microcontroller, mapping traversed cells and dynamically updating cost values to converge on the optimal path